

Project Design Phase

System Architecture Overview

The application follows a simple event-driven flow:

1. Student submits a request through the form
2. Data is saved into the custom table `u\_library\_book\_request`
3. Business Rules run automatically before the record is saved:
 - Generate a unique Request Number
 - Validate that the Issue Date is not in the past
4. Once the record is created, a Flow Designer workflow is triggered
5. The workflow sends an email notification to the librarian
6. All requests are visualized through Reports and a Dashboard

Table Design:

Table Name: `u\_library\_book\_request`

Table Label: Library Book Request

Application Scope: Library Book Request (custom scoped application)

Field List:

Student Name — `u\_student\_name` — String — Required

Student Email — `u\_student\_email` — Email — Required

Book Title — `u\_book\_title` — String — Required

Author — `u\_author` — String — Optional

Issue Date — `u\_issue\_date` — Date — Validated, cannot be in the past

Return Date — `u\_return\_date` — Date — Optional

Request Number — `u\_request\_number` — String — Auto-generated, read-only

Status — `u\_status` — Choice — Default: Pending

Librarian Notes — `u\_librarian\_notes` — String — Optional, filled by librarian

Form Layout Design

The form is organized into three logical sections:

Section 1 — Student Information

- Student Name
- Student Email

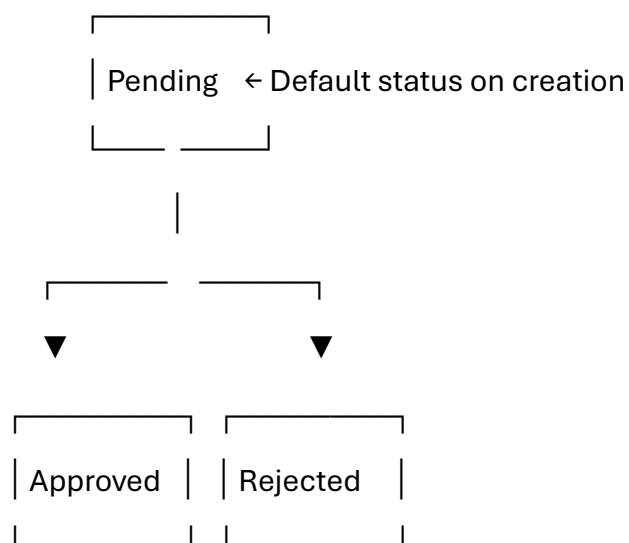
Section 2 — Book Details

- Book Title
- Author

Section 3 — Request Tracking

- Issue Date
- Return Date
- Request Number **(read-only)**
- Status
- Librarian Notes

Status Workflow Design



- A new request always starts as **Pending**
- The librarian manually changes it to **Approved** or **Rejected** after review

Notification Design (Flow Designer)

Trigger: Record Created on the `Library Book Request` table

Action: Send Email

Recipient: Librarian's email address

Subject: Includes the auto-generated Request Number

Body: Includes Request Number, Student Name, Student Email, Book Title, Author, Issue Date, Return Date, and Status — populated using Flow Designer data pills

Dashboard Design

Widget 1 — Pie Chart

- Source: Library Book Request table
- Grouped by: Status
- Purpose: Quick visual breakdown of Pending / Approved / Rejected requests

Widget 2 — List Report

- Source: Library Book Request table
- Columns: Request Number, Student Name, Book Title, Status, Issue Date
- Purpose: Detailed, scrollable view of all submitted requests

Role-Based Access Design

Admin Role (`x\_...\_librar\_0.admin`) — Full CRUD access

User Role (`x\_...\_librar\_0.user`) — Create, Read, Write only (no Delete)